

NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Third Semester B.Tech (Civil) (CBCS) Degree Examinations
 December 2023

CV2105-1 – CONCRETE TECHNOLOGY

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

* IS:456:2000 and IS:10262:2019 are permitted.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

1. The cement preferred in cold weather concreting is
 A) expansive B) low heat
 C) rapid hardening D) hydrophobic
2. In the manufacture of cement, the dry or wet mixtures of calcareous and argillaceous materials is burnt in
 A) silos B) rotary kiln
 C) feeder D) Tube mill
3. A dark solid nodular material formed by heating ground limestone and clay at a temperature of around 1300°C-1500° C is called
 A) clinker B) flakes
 C) slurry D) miller
4. Strength of concrete increases with
 A) increase in fineness of cement B) decrease in size of aggregates
 C) decrease in curing time D) increase in water cement ratio
5. Finely divided residue resulting from the combustion of powdered coal, transported by the flue gases, and collected by electrostatic precipitator is
 A) silica fume B) fly ash
 C) rice husk ash D) Cinder ash
6. Which type of aggregates give maximum strength to concrete?
 A) Angular aggregates B) Flaky aggregates
 C) Rounded aggregates D) Mixed aggregates
7. According to IS: 2386 part-IV, each layer is tamped _____ times in crushing test.
 A) 20 B) 30
 C) 25 D) 15
8. Which of the following can be used for curing of concrete?
 A) Ozone B) Chlorine
 C) Hydrogen peroxide D) Water
9. The ease with which the concrete can be mixed, placed and compacted in position in a homogeneous state is called.....
 A) durability B) workability
 C) bleeding D) segregation
10. The process of keeping concrete wet to enable it to attain full strength is known as
 A) curing B) immersion
 C) wetting D) quenching
11. The measurement of workability by Vee Bee consistometer test is expressed in
 A) dimensionless B) centimeter
 C) minutes D) seconds

12. Separation of cement paste from sand in the mortar allowing the water or cement paste to appear at the surface is called
- A) Honey combing B) segregation
C) bleeding D) scaling
13. The strength and durability of concrete does not depend on
- A) Size of aggregates B) Grading of aggregates
C) Moisture contents of aggregates D) Cement colour
14. Which among the following is not a type of Non-destructive testing?
- A) Visual testing B) Compression test
C) Ultrasonic testing D) Eddy current testing
15. Chloride mode of non entering into concrete
- A) cement B) admixtures
C) mixing water D) Fire
16. The durability of concrete is proportional to
- A) deterioration from environmental conditions B) internal disruptive forces
C) chemical attack D) Cement aggregate ratio
17. In M20 concrete M refers to _____
- A) Mix B) Minimum
C) Maximum D) Moderate
18. What is the density of normal concrete?
- A) 1800 kg/m³ B) 2400 kg/m³
C) 2800 kg/m³ D) 2000 kg/m³
19. What is the mix ratio for M10 grade of concrete?
- A) 1:2:4 B) 1:1.5:3
C) 1:1:2 D) 1:3:6
20. In making precast structural units for partition and wall lining purposes, the concrete used is
- A) Light weight B) Saw dust
C) High performance D) Pre-stressed

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) Explain the functions of cement ingredients. List any 3 field tests conducted for cement.	8	L*2	1	1
	b) Summarize the quality of mixing water parameters in concrete.	8	L2	2	1,7,8
2.	a) What are Bogue's compounds? Explain the properties of Bogue's compound.	8	L2	1	1,6,7
	b) Explain in detail the laboratory method to determine the impact value of coarse aggregate.	8	L2	2	1,6,7
3.	a) What is meant by hydration of cement? Explain the hydration process of cement with a neat flow chart.	8	L2	1	1,5,6
	b) What are chemical admixtures? How does the incorporation of accelerators and retarders affect the properties and performance of concrete.	8	L2	2	1,6,7
Unit – II					
4.	a) Define slump. Explain the detailed procedure for measurement of workability by slump test and show the different types of slump failures.	8	L2	3	1,5,6
	b) Explain the Ultrasonic pulse velocity to determine the strength of hardened concrete.	8	L2	4	1,5,6
5.	a) Summarize segregation and bleeding. What are the methods adopted to prevent segregation and bleeding.	8	L2	3	1,5,6
	b) Define gel space ratio. Determine the gel/space ratio and the theoretical strength of a sample of concrete made with 500 gm, of cement with 0.45 w/c ratio, on full hydration and at 55 percent hydration.	8	L5	4	1,2,6,7

- | | | | | | | |
|----|----|---|---|----|---|-------|
| 6. | a) | Explain the various factors affecting workability of concrete. | 8 | L2 | 3 | 1,5,6 |
| | b) | Explain the influence of water/cement ratio and maturity concept on the strength of concrete. | 8 | L2 | 4 | 1,5,6 |

Unit – III

- | | | | | | | |
|----|--|--------------------------|----|----|---|-----------|
| 7. | Explain the following types of special concrete with respect to materials, properties, and applications: | | | | | |
| | i) | High density concrete | | | | |
| | ii) | Self-compacting concrete | 16 | L2 | 5 | 1,5,6,7 |
| 8. | Design a concrete mix by BIS method with the following data | | | | | |
| | Grade designation: M30 | | | | | |
| | Maximum nominal size of aggregate: 20mm (crushed angular) | | | | | |
| | Fine aggregates: Conforming to grading zone II | | | | | |
| | Target slump: 100–125 mm | | | | | |
| | Degree of supervision: Good | | | | | |
| | Type of exposure: Mild | | | | | |
| | Method of placement: Pumping | | | | | |
| | Chemical admixture type: Plasticizer | | | | | |
| | Specific gravity of cement: 3.05 | | | | | |
| | Specific gravity of admixture: 1.21 | | | | | |
| | Specific gravity of coarse aggregate: 2.75 | | | | | |
| | Specific gravity of fine aggregate: 2.65 | | | | | |
| | Assume any other data if missing. Also calculate the quantity of cement, sand, coarse aggregate, and water required per cubic meter of concrete. | | 16 | L6 | 5 | 1,2,6,7,8 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
